Two-Stage Product Search Ranker

Hybrid retrieval and a cascade of rankers for the Shopping Queries (ESCI) dataset

Task

Given a shopping query, rank products from a 482k catalog so that exact matches come first. The whole search must answer within 100 ms. Every labelled query-product pair carries one of four ESCI labels, and each label has a gain.

	Label	Gain	Share
E	Exact: matches the query	1.0	43.5%
S	Substitute: some detail differs	0.1	35.1%
C	Complement: goes with it	0.01	4.5%
I	Irrelevant	0	16.9%

Table 1: ESCI labels on the US English data.

Data

We use the US English part of task 1: 29.8k queries and 601k labelled pairs. Queries are split, never rows, so no query sits in two splits. The cross-encoder trains on its own queries, so LightGBM never learns from scores the cross-encoder made on its own training data.

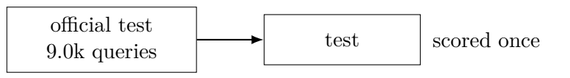

Figure 1: Query-level split.

Metrics

With g_i the gain of the product at rank i,

$$\mathrm{DCG@}k = \sum_{i=1}^{k} \frac{g_i}{\log_2(i+1)}, \qquad \mathrm{NDCG@}k = \frac{\mathrm{DCG@}k}{\mathrm{IDCG@}k}.$$

Full-list NDCG takes k as each query's whole labelled list, as the KDD Cup leaderboard did. **NDCG@10** looks only at what a shopper sees, and is much stricter. **Recall@k** is the share of Exact products that retrieval finds in its top k of the whole catalog.

Latency

500 test queries replayed one at a time through the HTTP API, on an Apple M4 Pro laptop.

Stage	p50 (ms)	p95 (ms)
Encode query	4.7	5.3
Retrieve 200 candidates	3.0	4.8
Features for 200	14.7	18.2
Light ranker	0.6	0.7
Cross-encoder on top 50	19.2	22.2
Full ranker	0.6	0.8
Total, including HTTP	**44.7**	**49.9**

Table 2: Per-stage latency.

System

Stage one recalls candidates cheaply from the whole catalog. Stage two spends more compute on fewer products: a light ranker sorts all 200 candidates, and only its top 50 reach the cross-encoder.

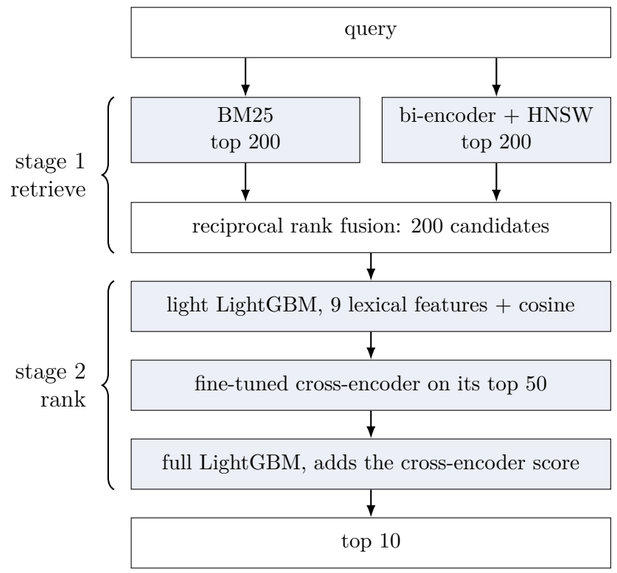

Figure 2: Online serving cascade.

Cross-encoder

A bi-encoder embeds query and product separately. A cross-encoder reads them together, so every query token attends to every title token. We fine-tune `ms-marco-MiniLM-L6` (22M parameters) for one epoch on 8.3k ce_train queries, about 5 minutes on the laptop GPU. The target is the ESCI gain itself, so Substitutes count as partly relevant.

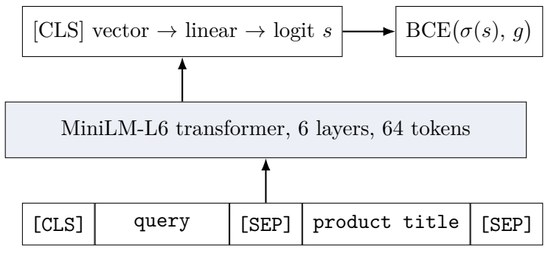

Figure 3: Cross-encoder input and training target $g \in \{1, 0.1, 0.01, 0\}$.

Cross-encoder	NDCG@10	Full NDCG
Zero-shot, alone	0.719	0.836
Fine-tuned, alone	0.750	0.853
Fine-tuned, inside full LightGBM	**0.759**	**0.858**

Table 3: Effect of fine-tuning, test split.

Results

Each idea, added in turn and scored on the untouched official test split. On the competition's full-list metric the system reaches 0.858, above the official 0.83 baseline.

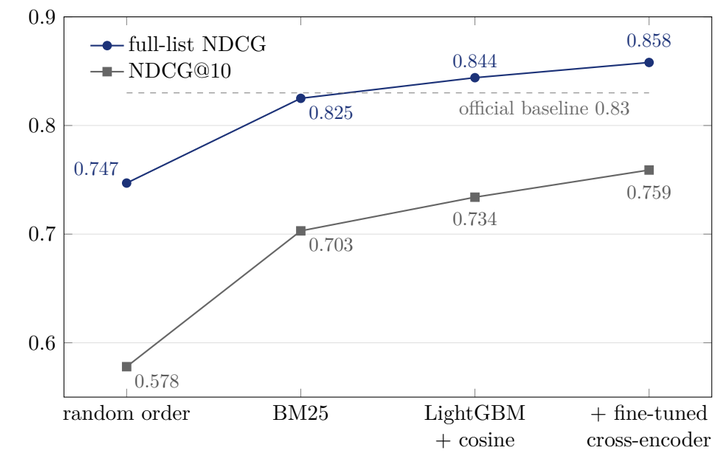

Figure 4: NDCG on the official test split. The dashed line is the official baseline on the full-list metric.

Lexical features stop helping around 0.725 NDCG@10. What is left is mostly telling Exact from Substitute when the words overlap, which needs a model that reads both texts.

Retrieval, 482k products	R@100	R@200
BM25	0.578	0.652
Bi-encoder + HNSW	0.518	0.598
Hybrid, RRF	**0.628**	**0.697**

Table 4: Recall of Exact products.

Searching the whole catalog end to end, the top 10 scores 0.372 NDCG@10, against 0.317 for BM25 alone. Unlabelled products count as 0, so this is a lower bound.

Monitoring

Requests are logged and compared with a reference by population stability index (PSI). Above 0.25 counts as drift.

Traffic	Query length	Top score
Replayed test queries	0.001	0.059
Spanish queries	0.093	**1.086**

Table 5: PSI on an input and an output signal.

Summary

- BM25 and dense retrieval miss different products, so fusing them finds 5 points more Exact products.
- The fine-tuned cross-encoder gives the largest single gain, and takes 68% of the full ranker's split gain.
- Running it only on the light ranker's top 50 keeps p95 latency at 50 ms.
- An input-only monitor would have missed the Spanish traffic. The output score caught it.